MATH2241 Introduction to Mathematical Analysis
Review Session 1: Practice Problems and Solutions

STA: Xue Yingcong

Oct 5, 2026

Problem 1. Let A be a non-empty subset of $\mathbb{R}$ and let α be an upper bound of A. Suppose that for every $n \in \mathbb{N}$, there exists $a_n \in A$ such that $\alpha - \frac{1}{n} < a_n$. Prove that $\alpha = \sup A$.

Answer. Since α is an upper bound of A, we must have $\alpha \geq \sup A$.

Assume for contradiction that $\alpha > \sup A$. Let $r := \alpha - \sup A > 0$. Choose an $n \in \mathbb{N}$ such that $n > \dfrac{1}{r}$. Then $r > \dfrac{1}{n}$ and thus there exists some $a_n \in A$ such that

$$\sup A = \alpha - r < \alpha - \frac{1}{n} < a_n.$$

This contradicts the fact that $\sup A$ is an upper bound of A. Therefore, $\alpha = \sup A$.

Problem 2. Give an example of each of the following, or prove that such a situation is impossible:

(a) sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n\}$ converges, $\{b_n\}$ is unbounded, and $\{a_n b_n\}$ converges.

(b) sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n\}$ is unbounded, $\{b_n\}$ converges, and $\{a_n - b_n\}$ is bounded.

(c) sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n\}$ converges, $\{b_n\}$ diverges, and $\{a_n + b_n\}$ converges.

(d) sequences $\{a_n\}$ and $\{b_n\}$, which both diverge, but $\{a_n + b_n\}$, $\{a_n b_n\}$, and $\{a_n / b_n\}$ all converge.

Answer.

(a) Take $\{a_n\} = \{0, 0, 0, 0, \ldots\}$ and $\{b_n\} = \{1, 2, 3, 4, \ldots\}$. Then $\{a_n\}$ converges to 0, $\{b_n\}$ is unbounded (not bounded above), and $\{a_n b_n\}$ converges to 0.

(b) *Impossible.* Since $\{b_n\}$ converges, $\{b_n\}$ is bounded, so there exists $M_1 > 0$ such that $|b_n| \leq M_1$ for all $n \in \mathbb{N}$. Similarly, there exists $M_2 > 0$ such that $|a_n - b_n| \leq M_2$ for all $n \in \mathbb{N}$. Since $a_n = (a_n - b_n) + b_n$, we have for any $n \in \mathbb{N}$,

$$|a_n| = |(a_n - b_n) + b_n| \leq |a_n - b_n| + |b_n| \leq M_1 + M_2,$$

and hence $\{a_n\}$ must also be bounded.

(c) *Impossible.* If $\{a_n + b_n\}$ and $\{a_n\}$ converge, then $\{b_n\} = \{(a_n + b_n) - a_n\}$ also converges.

(d) Take $\{a_n\} = \{1, -1, 1, -1, \ldots\}$ and $\{b_n\} = \{-1, 1, -1, 1, \ldots\}$. Then both $\{a_n\}$ and $\{b_n\}$

diverge. However, the sequences

$$\{a_n + b_n\} = \{0, 0, 0, 0, \ldots\},$$
$$\{a_n b_n\} = \{-1, -1, -1, -1, \ldots\},$$
$$\{a_n / b_n\} = \{-1, -1, -1, -1, \ldots\}$$

are all convergent.

Problem 3. Let $a > 1$. Define a sequence $\{x_n\}$ by

$$x_1 = a, \qquad x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right) \quad \text{for any } n \in \mathbb{N}.$$

(a) Prove by induction that $x_n > \sqrt{a}$ for any $n \in \mathbb{N}$.

(b) Prove that $\{x_n\}$ is decreasing. Hence show that $\{x_n\}$ converges. Find also the limit.

Answer.

(a) First of all, we have $x_1 = a > \sqrt{a}$ as $a > 1$. Assume that $x_k > \sqrt{a}$. Then we have $(x_k - \sqrt{a})^2 > 0$. Expanding this gives

$$\begin{aligned}(x_k - \sqrt{a})^2 > 0 &\implies x_k^2 - 2x_k\sqrt{a} + a > 0 \implies x_k^2 + a > 2x_k\sqrt{a} \\ &\implies \frac{x_k^2 + a}{2x_k} > \sqrt{a} \\ &\implies x_{k+1} > \sqrt{a}.\end{aligned}$$

Thus, by induction, we have shown that $x_n > \sqrt{a}$ for any $n \in \mathbb{N}$.

(b) By (a), we have $x_n^2 > a$ for any $n \in \mathbb{N}$. Adding x_n^2 on both sides yields $2x_n^2 > x_n^2 + a$. Hence we have

$$x_n > \frac{x_n^2 + a}{2x_n} = x_{n+1}.$$

It follows that the sequence $\{x_n\}$ is decreasing. Since $\{x_n\}$ is decreasing and bounded below (by part (a), $x_n > \sqrt{a}$ for all n), the Monotone Convergence Theorem implies that $\{x_n\}$ converges. Let $L = \lim_{n\to\infty} x_n$. Then L must satisfy

$$L = \frac{L^2 + a}{2L}, \qquad \text{that is,} \qquad L^2 = a.$$

Since all terms of the sequence are positive, we must have $L = \sqrt{a}$.

Problem 4. Let $x_1 := a > 0$ and $x_{n+1} := x_n + 1/x_n$ for $n \in \mathbb{N}$. Determine whether (x_n) converges or diverges.

Answer. Assume that (x_n) converges. Then (x_n) is bounded, hence there exists some $u \in \mathbb{R}$ such that $|x_n| \le u$ for all n. As $|x_1| = x_1 = a \le u$ and $a > 0$, we must have $u > 0$. Then $|x_{n+1} - x_n| = \left|\frac{1}{x_n}\right| \ge \left|\frac{1}{u}\right|$ for all n. By taking $\epsilon = \left|\frac{1}{u}\right| > 0$, for any $N \in \mathbb{N}$ there exist $m = N + 1$, $n = N$ such that $|x_m - x_n| = \left|\frac{1}{x_n}\right| \ge \epsilon$, hence (x_n) is **not** Cauchy and hence does **not** converge, a contradiction. Therefore, (x_n) diverges.

Problem 5. Let $0 < \alpha < 1$ and let $\{a_n\}$ be a sequence which satisfies

$$|a_{n+2} - a_{n+1}| \leq \alpha |a_{n+1} - a_n| \qquad \text{for all } n \in \mathbb{N}.$$

Prove that $\{a_n\}$ is a Cauchy sequence (and hence converges).

Answer. The given condition implies that

$$|a_{n+2} - a_{n+1}| \leq \alpha^n |a_2 - a_1| \qquad \text{for all } n \in \mathbb{N}.$$

Note that if $a_1 = a_2$, then the above suggests that $\{a_n\}$ is a constant sequence and hence must be a Cauchy sequence. Thus we may assume that $a_1 \neq a_2$.

If $n > m \geq 2$, then

$$\begin{aligned} |a_n - a_m| &= |(a_n - a_{n-1}) + (a_{n-1} - a_{n-2}) + \cdots + (a_{m+1} - a_m)| \\ &\leq |a_n - a_{n-1}| + |a_{n-1} - a_{n-2}| + \cdots + |a_{m+1} - a_m| \\ &\leq (\alpha^{n-2} + \cdots + \alpha^{m-1})|a_2 - a_1| \\ &= \alpha^{m-1} \left(\frac{1 - \alpha^{n-m}}{1 - \alpha} \right) |a_2 - a_1| \\ &< \alpha^{m-1} \left(\frac{1}{1 - \alpha} \right) |a_2 - a_1| = C\alpha^{m-1}, \end{aligned}$$

where $C = \left(\dfrac{1}{1 - \alpha} \right) |a_2 - a_1|$ is a positive constant. Now since $0 < \alpha < 1$, the sequence $\{\alpha^m\}$ converges to 0. Thus for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for any $m \in \mathbb{N}$, if $m \geq N$,

$$\alpha^m < \frac{\alpha}{C}\epsilon, \qquad \text{or equivalently,} \qquad \alpha^{m-1} < \frac{1}{C}\epsilon.$$

It follows that for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for any $n, m \in \mathbb{N}$, if $n > m \geq N$, then $|a_n - a_m| < \epsilon$. This shows that $\{a_n\}$ is Cauchy, and hence converges.

Problem 6. Determine whether each of the following series converges. Justify your answers.

(a) $\displaystyle\sum \frac{n^2}{n!}$

(b) $\displaystyle\sum \left[\sqrt{n+1} - \sqrt{n}\right]$

(c) $\displaystyle\sum \frac{n!}{n^n}$

(d) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{\sqrt{n} \log n}$

Answer.

(a) Let $a_n := \dfrac{n^2}{n!}$. Notice that $a_n \neq 0$ for all n.

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{n+1}{n^2} \right| = 0.$$

By the Ratio Test, as $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = 0 < 1$, the series converges.

(b) Let $s_n := \sum_{i=1}^{n} \left[\sqrt{i+1} - \sqrt{i}\right]$. This sum telescopes, giving $s_n = \sqrt{n+1} - 1$. Therefore,

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} \left(\sqrt{n+1} - 1\right) = \infty,$$

so (s_n) diverges, and hence the series diverges. (Note that the n-th term test alone does not settle this, since $\sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1}+\sqrt{n}} \to 0$.)

(c) Let $a_n := \dfrac{n!}{n^n}$. Notice that $a_n \neq 0$ for all n. Simplifying,

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!/(n+1)^{n+1}}{n!/n^n} = \frac{(n+1)\,n^n}{(n+1)^{n+1}} = \left(\frac{n}{n+1}\right)^n \longrightarrow e^{-1}.$$

By the Ratio Test, as $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = e^{-1} < 1$, the series converges.

(d) For sufficiently large n, $1 \leq \log n \leq \sqrt{n}$, hence

$$\frac{1}{\sqrt{n}\log n} \geq \frac{1}{\sqrt{n}\cdot\sqrt{n}} = \frac{1}{n}.$$

Since $\sum \frac{1}{n}$ diverges (the harmonic series), by the Comparison Test $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}\log n}$ also diverges.

Problem 7.

(a) Suppose the series $(s_n = \sum_{k=1}^{n} a_k)$ is absolutely convergent. Prove that the series $\left(t_n = \sum_{k=1}^{n} a_k^3\right)$ is convergent.

(b) More generally, what can be said about $\sum a_n^k$ for other integers k?

Answer.

(a) Since (s_n) is absolutely convergent, it follows from the Cauchy criterion that if $\epsilon^{1/3} > 0$ is given, then there exists $N_\epsilon \in \mathbb{N}$ such that

$$|a_m| + |a_{m-1}| + \cdots + |a_{n+1}| < \epsilon^{1/3}$$

for all $m > n \geq N_\epsilon$.

To finish, it suffices to prove that (t_n) is absolutely convergent by the Cauchy criterion. Given $\epsilon > 0$, take the natural number $N_\epsilon \in \mathbb{N}$ above, and then one obtains

$$|a_m|^3 + |a_{m-1}|^3 + \cdots + |a_{n+1}|^3 \leq \left(|a_m| + |a_{m-1}| + \cdots + |a_{n+1}|\right)^3 < \left(\epsilon^{1/3}\right)^3 = \epsilon$$

for all $m > n \geq N_\epsilon$.

(b) The argument in part (a) did not really use that the exponent is 3: for any integer $k \geq 1$, the same proof goes through verbatim (with $\epsilon^{1/3}$ replaced by $\epsilon^{1/k}$), using the inequality

$$x_1^k + x_2^k + \cdots + x_r^k \leq (x_1 + x_2 + \cdots + x_r)^k$$

valid for any real numbers $x_1, \ldots, x_r \geq 0$ and any $k \geq 1$ (since $(x_1 + \cdots + x_r)^k = \sum_i x_i(x_1 + \cdots + x_r)^{k-1} \geq \sum_i x_i \cdot x_i^{k-1} = \sum_i x_i^k$, as $x_i \leq x_1 + \cdots + x_r$). Hence **for every integer** $k \geq 1$, $\sum a_n^k$ is absolutely convergent, and therefore convergent.

For non-positive integers the conclusion fails in general. If $k = 0$, then $a_n^k = 1$ for every n with $a_n \neq 0$, so $\sum a_n^0$ diverges whenever infinitely many a_n are non-zero. If $k < 0$, then since $\sum a_n$ converges we must have $a_n \to 0$; along the terms with $a_n \neq 0$, $a_n^k = 1/a_n^{|k|} \to \infty$, so the terms of $\sum a_n^k$ do not tend to 0 and the series diverges by the n-th term test (terms with $a_n = 0$ make a_n^k undefined in any case). Thus **for every integer** $k \leq 0$, $\sum a_n^k$ diverges or is not well-defined, in general.

Problem 8.

(a) Find a series $\sum a_n$ which diverges by the Root Test but for which the Ratio Test gives no information.

(b) Find a series $\sum a_n$ which converges by the Root Test but for which the Ratio Test gives no information.

Answer.

(a) Let $a_n = 2^{\,n+(-1)^n}$ for $n \in \mathbb{N}$.

Root Test. Since $\dfrac{(-1)^n}{n} \to 0$,

$$\sqrt[n]{a_n} = 2^{\,1+\frac{(-1)^n}{n}} \longrightarrow 2^1 = 2 > 1,$$

so the Root Test shows that $\sum a_n$ diverges.

Ratio Test. Since the exponent satisfies $\big(n{+}1 + (-1)^{n+1}\big) - \big(n + (-1)^n\big) = 1 - 2(-1)^n$,

$$\frac{a_{n+1}}{a_n} = 2^{\,1-2(-1)^n} = \begin{cases} 8, & n \text{ odd}, \\ \dfrac{1}{2}, & n \text{ even}. \end{cases}$$

This alternates forever between 8 and $\dfrac{1}{2}$, so $\lim_{n\to\infty} a_{n+1}/a_n$ does not exist and the Ratio Test gives no information.

(b) Let $a_n = 2^{(-1)^n - n}$ for $n \in \mathbb{N}$.

Root Test. Since $\dfrac{(-1)^n}{n} \to 0$,

$$\sqrt[n]{a_n} = 2^{\,\frac{(-1)^n}{n} - 1} \longrightarrow 2^{-1} = \frac{1}{2} < 1,$$

so the Root Test shows that $\sum a_n$ converges (absolutely).

Ratio Test. Since the exponent satisfies $\big((-1)^{n+1} - (n{+}1)\big) - \big((-1)^n - n\big) = -2(-1)^n - 1$,

$$\frac{a_{n+1}}{a_n} = 2^{\,-2(-1)^n - 1} = \begin{cases} 2, & n \text{ odd}, \\ \dfrac{1}{8}, & n \text{ even}. \end{cases}$$

This alternates forever between 2 and $\dfrac{1}{8}$, so the limit does not exist and the Ratio Test gives no information.